

UK Lunabotics 2027



Challenge Overview and Registration Requirements

Registration deadline: Friday 30 October 2026, 23:59 (UK time)

Competition: 6–8 July 2027, The Open University, Milton Keynes

uklunabotics.co.uk | uklunabotics@open.ac.uk

Version 2.0 – 23 September 2026

Welcome to UK Lunabotics 2027

Thank you for your interest in UK Lunabotics. Whether this is your team's first engineering competition or you have entered similar challenges before, you are very welcome. You do not need previous experience of robotics competitions or space engineering to take part. What you need is a group of enthusiastic students, a supportive member of staff and the commitment to see a project through over the academic year.

This document explains what the challenge is, what taking part involves, who can enter and how to apply. If anything is unclear, please email us at uklunabotics@open.ac.uk. We are happy to answer questions at any stage, including before you apply.

What you are signing up for

By registering, your team commits to **designing and building a lunar construction rover using a systems engineering process**, and to **bringing that rover to compete at The Open University campus in Milton Keynes** in July 2027. Over the 2026/27 academic year your team will:

1. **Plan your project.** Form your team, find an Academic Supervisor and write a short Project Management Plan (PMP). This is your application.
2. **Design your rover using systems engineering.** Work through the same review stages NASA uses: a System Requirements Review (SRR), a Preliminary Design Review (PDR) by 29 January 2027 and a Critical Design Review (CDR) by 12 March 2027. You run these reviews yourselves, and we encourage you to invite UK Lunabotics organisers and our industry partners to give feedback.
3. **Build and test your rover.** Manufacture, integrate, test and improve your design.
4. **Show us it works.** Submit a short Proof of Life video by 14 May 2027 showing your rover moving material and its emergency stop working.
5. **Compete.** Bring your rover to our lunar arena at The Open University, Milton Keynes, on 6–8 July 2027 and build a berm from simulated lunar regolith.

UK Lunabotics 2026

UK Lunabotics ran for the first time in 2026. Seven teams took part in the challenge, and four of them brought their rovers to the competition at The Open University in Milton Keynes. The winning team has subsequently been invited as the UK entry to the international version of this competition, to be held in the USA in late 2026. We are very grateful to every team, supervisor, judge and partner who helped make the first year a success, and we look forward to welcoming new and returning teams in 2027.

UK Lunabotics 2027 at a Glance

What	A year-long engineering challenge. Student teams design, build and test a prototype lunar construction rover using systems engineering, then compete at an on-site event.
The task	Drive (by remote control or autonomously) across a lunar sandbox with obstacles, collect regolith simulant from an excavation zone and build a berm in a target area.
Who can enter	Teams of students enrolled at UK universities and further/higher education colleges (including degree apprentices), supported by an Academic Supervisor from their institution. See 'Who Can Enter'.
Team size	At least two undergraduate or further education students. There is no maximum team size, but space for team members at the competition event is limited to around 10 members per team – exact number will be communicated.
Cost	Free to enter. Teams fund their own rover, travel and accommodation.
Where	The Open University campus, Milton Keynes.
How to apply	Email a Statement of Support and a Project Management Plan (max. 5 pages) to uklunabotics@open.ac.uk by Friday 30 October 2026, 23:59 (UK time) . See 'How to Apply'.
Rules and updates	The challenge rulebook and news are published at uklunabotics.co.uk .
Questions	uklunabotics@open.ac.uk

Key dates

Date	Milestone
Fri 30 October 2026, 23:59	Registration deadline. Email your Statement of Support and Project Management Plan.
By Fri 20 November 2026	Teams notified whether they have a place.
Wednesday 2nd December	Information day at The Open University, Milton Keynes; Remote connection will be possible but in-person attendance is encouraged.
By Fri 29 January 2027	Preliminary Design Review (PDR) held by each team.
By Fri 12 March 2027	Critical Design Review (CDR) held by each team.
Fri 19 March 2027	Down selection to 8 teams
By Fri 14 May 2027	Proof of Life video due.
Fri 21 May 2027	Places at the competition confirmed to teams.
Tue 6 July 2027	Arrival at The Open University for rover assembly and test
Wed 7 – Thu 8 July 2027	UK Lunabotics Competition at The Open University.

Background

The Lunabotics Challenge is a NASA-led engineering challenge in which university student teams from across the United States build and test prototype off-world construction robots. First run in 2010, it supports NASA's plans for sustained exploration of the Moon, and later Mars. Teams work on the kind of problems that future lunar missions will face: digging and moving hard, abrasive regolith (lunar soil), travelling across uneven terrain with obstacles, and operating reliably with little or no direct human control. At the same time, students develop the engineering and teamwork skills needed by the next generation of space engineers.

The challenge has since expanded beyond the USA to countries including India and Australia. **UK Lunabotics is run by The Open University and is based on NASA's Lunabotics Challenge.** Caterpillar, the main sponsor of the US challenge, together with NASA's Solar System Exploration Research Virtual Institute (SSERVI), supported bringing the competition to the UK. The US competition organisers and Caterpillar UK provide input to UK Lunabotics. UK Lunabotics is not run by NASA, and teams should not describe themselves as being affiliated with or sponsored by NASA.

Objectives and Aims

The main objective of UK Lunabotics is for teams to apply the NASA systems engineering process to design, develop, build and test a prototype lunar robot that supports simulated lunar construction. Your robot must be able to cross challenging lunar terrain and build a berm (a raised bank) from regolith simulant.

Why berms matter

On the Moon, berms built from local regolith could protect astronauts and equipment on missions such as Artemis. They could provide:

- protection from the blast and plume ejecta (fast-moving dust and rock) thrown out when spacecraft land and launch
- radiation shielding for habitats, propellant storage and nuclear power systems
- thermal shading for cryogenic propellant tanks
- protection for other critical infrastructure on the lunar surface.

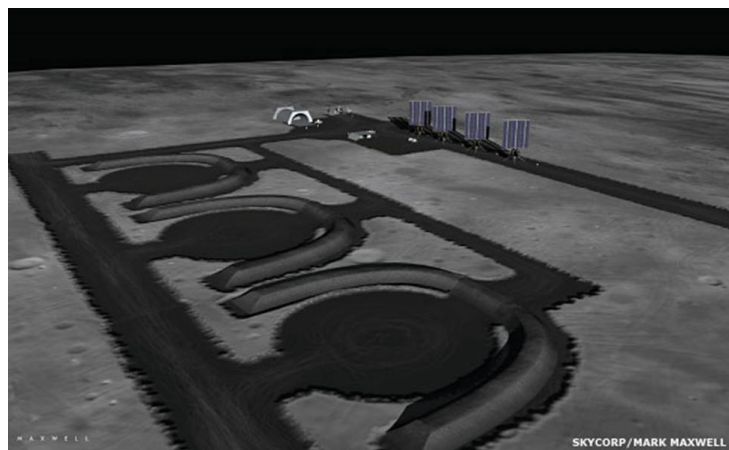


Figure 1. Concept of lunar landing pads with berms (Source: NASA / SkyCorp)



Figure 2. Concept of lunar landing pads (Source: NASA)

What your team will gain

UK Lunabotics gives students practical experience of systems, mechanical, electrical and software engineering, including:

- writing software for remote and autonomous construction operations
- designing robot frames, mobility systems, sensors and electronics
- using a range of manufacturing processes
- stress-testing mechanical components
- troubleshooting, prototyping and improving a design through iteration
- operating hardware remotely and autonomously.

Along with project management, technical documentation, teamwork and outreach, these are skills that employers value and that will prepare you for a career in engineering.

The Challenge

Your rover's mission is to build a berm that protects astronauts and habitat modules from plume ejecta when spacecraft land and take off on the other side of it. The berm acts as a wall between the landing pad and the habitat. In general, a larger berm gives more protection.

In the competition arena, your rover must:

- cross a lunar sandbox containing obstacles, operated either by remote control or autonomously
- collect regolith simulant from a designated excavation zone
- carry it to a construction zone and build a berm within the target area (1.5 m × 0.9 m).

During each run, your team operates the rover from an operations shed, where you can see video feeds from cameras around the arena. You will not be able to see the arena directly. The full arena layout, dimensions and run rules are in the challenge rulebook at uklunabotics.co.uk.

Design reviews (PDR and CDR)

Design reviews are checkpoints where your team presents its design, checks it against your requirements and decides whether you are ready to move on to the next stage. Your team plans and runs its own reviews, and there is nothing to submit to us. Your PDR must take place by 29 January 2027 and your CDR by 12 March 2027.

Your team may want to hold a System's Requirements Review after iterating on feedback from your PMP.

We strongly encourage you to invite UK Lunabotics organisers and our industry partners to your PDR and CDR to give independent feedback. If you would like outside expertise at a review, the organising committee can help you find it. Email us well in advance of your review date.

Proof of Life video

The Proof of Life video shows that your rover works and that you are on track to compete. It must show your rover driving, and it must show the rover's emergency stop being used and working. After reviewing the videos, we will confirm places at the on-site competition by 21 May 2027.

Systems Engineering Paper (optional)

A Systems Engineering Paper is not required for UK Lunabotics 2027, but we encourage it. In a Systems Engineering Paper, your team describes how it applied systems engineering to the project: your requirements, design reviews, trade-offs, testing, and how your plans changed. Writing one is excellent practice for professional engineering work and a useful record to pass on to future teams. The systems engineering paper may become a component of scoring in future years.

Scoring

At the competition, every rover must first pass the inspections. It is then scored in four areas. The full scoring formulae and rules are in the challenge rulebook at uklunabotics.co.uk.

Before you compete: inspections (pass/fail)

Each rover must pass a mechanical inspection and a communications check before it is allowed into the arena. These are not scored, but a rover cannot compete until it passes.

Scored area	What is measured
Berm Productivity – normalised for robot mass (BP Mass)	The arena is scanned before and after your run to measure the volume of the berm. Only berm volume inside the 1.5 m × 0.9 m target area counts. There is no restriction on the shape, height or number of berms. The result is normalised for your rover's mass, so lighter rovers are rewarded.
Berm Productivity – normalised for energy (BP Energy)	The same berm volume measurement, normalised for the electrical energy your rover uses. Energy use must be shown on a commercial off-the-shelf (COTS) electronic data logger and checked by a judge.
Bandwidth use	Arena cameras are available to help you operate your rover remotely. Before each run, you choose how many to use.
Autonomy	Autonomy tasks of different difficulty are worth between 50 and 600 points. You must tell the judges when you start and finish an autonomy task so it can be verified. The tasks are listed in the rulebook.

The Competition Event

The UK Lunabotics Competition takes place over three days, 6–8 July 2027, at The Open University campus in Milton Keynes. The event includes rover inspections, practice and competition runs in our lunar arena, and ends with awards. A detailed programme will be sent to teams with places before the event.

Presentation and Demonstration (P&D)

During the event, each team gives a short presentation to the judges about its rover design, how the team worked and each member's role in the project.

Attending the event

- Space for team members at the event is limited. We will confirm how many members from each team can attend nearer the time.
- Your Academic Supervisor must attend and stay on site for the whole event.
- Entry is free. Teams are responsible for their own travel, accommodation and rover shipping costs, so include these in your budget.

Support from the organisers

- Email uklunabotics@open.ac.uk with questions at any time. No question is too basic, especially for teams new to this kind of competition.
- Teams will be given access to a Open University online hub where they can communicate with other teams, privately within their team, and with organising staff and other members of the competition team. Access will be given on successful application to the competition.
- The organising committee can help you find reviewers and external expertise for your design reviews.
- The rulebook, updates and resources are published at uklunabotics.co.uk.

Who Can Enter

Team requirements

- All team members must be students enrolled at a UK university or further/higher education college (including degree apprentices) during the 2026/27 academic year. You may be enrolled for any part of the year.
- Teams must include at least two undergraduate or further education students. Postgraduate students are welcome, and there is no limit on their number.

- There is no maximum team size. However, space at the competition event is limited (see 'The Competition Event').
- Teams from more than one institution are allowed (for example, two institutions working together as one team).
- All students must be 18 or over before attending any on-site event.
- Each team must have an **Academic Supervisor**: a member of staff employed by the registered institution, aged 21 or over, who supports the team throughout the project and attends the whole competition event.

Doing the work

- All team members should be actively involved in designing and building the rover, with clear roles that you will describe at the Presentation and Demonstration.
- Students must do all of the project work, including design, construction, written documents, presentations and competition preparation. Professionals (including staff and technicians) may help only with training or safety, under student direction.

Code of conduct

We want UK Lunabotics to be safe, friendly and fair for everyone. We expect all team members and supervisors to:

- treat other teams, judges, volunteers and staff with respect, and show good sportsmanship
- follow the competition rules and the instructions of event staff, especially on safety
- be honest about their team, including not claiming to be affiliated with or sponsored by NASA, and not impersonating NASA or partner staff
- use NASA, sponsor or other organisations' logos and trademarks only with permission.

Serious or repeated breaches of this code, or failure to meet the requirements in this document, may lead to an individual or team being disqualified.

How to Apply

Application deadline: Friday 30 October 2026, 23:59 (UK time)

Email both the statement of support and PMP documents as PDFs to uklunabotics@open.ac.uk. Applications must be sent by the Academic Supervisor or the team lead.

What to submit

Document	File name	See
Statement of Support	SOS-Institution Name	Statement of Support section
Project Management Plan (max. 5 pages)	PMP-Institution Name	Project Management Plan section

- Submit each document as a PDF.
- Name your files as shown, using your institution's official name rather than your team name, for example **SOS-The Open University** and **PMP-The Open University**. Files that do not follow this naming may be rejected.
- For teams from more than one institution, use the lead institution's name.

How applications are assessed

We check every application for completeness and compliance, then score and rank the Project Management Plans. There is no fixed number of places. The number of teams accepted depends on the quality of applications and the capacity of the event. If we receive more good applications than we have places for, places are offered in order of PMP score.

We will email you by 20 November 2026 to tell you whether your team has a place. Submitting an application does not guarantee a place.

Statement of Support

Your application must include a Statement of Support from your Academic Supervisor, confirming that they are willing to supervise and support the team through every stage of UK Lunabotics. We cannot accept teams without an Academic Supervisor. The Academic Supervisor should also sign the cover of each document you submit, to show that they have seen and approved it.

The following statement should appear on institution letterhead, signed by the Academic Supervisor and team members:

Academic Supervisor:

Institution:

I support the concepts and methods by which the students plan to compete in UK Lunabotics 2027. I will support the team members to complete all project requirements and meet deadlines, and I will let the organisers know promptly if the team needs to withdraw.

Signature: _____ Date: _____

	Name (print)	Signature	Role (Supervisor / Student)
1			
2			
3			
4			
5			
6			
7			

Add more rows if needed.

Project Management Plan

A Project Management Plan (PMP) is a standard systems engineering document. It sets out the timeline, objectives, budget and technical targets for a project at the planning stage. For UK Lunabotics, the PMP is your main application document. It shows us that your team has thought through what the project will involve, and it is used to allocate places if we have more applications than places.

Your PMP can be **no more than 5 pages** long. It is a starting plan, not a fixed commitment: we expect your plans to change as the project progresses. If you are new to systems engineering, the resources in the Project Management Plan Guidance section and the Glossary at the end of this document will help.

Your PMP should contain the following four sections.

1. Initial Project Schedule

Provide a Gantt chart or equivalent showing your project's major dates and events, including at least the items below. Discuss these only as needed.

- a) **Start date.**
- b) **Completion date:** the date the project ends after the competition, when you hand the rover over to next year's team, dispose of it or otherwise decommission it.
- c) **Dates of your major design reviews,** including at least:
 - System Requirements Review (SRR) (optional)
 - Preliminary Design Review (PDR), by 29 January 2027
 - Critical Design Review (CDR), by 12 March 2027
 - any other reviews you find useful.
- d) **Delivery dates to UK Lunabotics,** including at least:
 - your Proof of Life video (due 14 May 2027)
 - your Systems Engineering Paper, if you plan to write one (optional).
- e) **Budget and technical milestones:** key dates for your cost budget and Technical Performance Measures, as identified in sections 2 and 4 of your PMP. You may also include major systems engineering activities.
- f) **Managing the schedule:** how you will keep the schedule up to date during the project (how often you will review it) and how you will respond if you fall behind or get ahead.

2. Initial Project Cost Budget

Estimate the total cost of the project, including all likely costs. Provide a table of major budget categories and items, including at least the items below. Discuss these only as needed.

- a) **A breakdown of your total cost estimate** (the items should add up to your total), including:
 - cost estimates for each element of your early System Hierarchy (include the hierarchy itself; at this stage it may only have two levels)
 - labour costs, if any
 - material costs for building the rover and producing your UK Lunabotics deliverables
 - travel and accommodation costs for attending the competition at The Open University, Milton Keynes, on 6–8 July 2027, and any costs of transporting your rover.
- b) **Critical budget milestones and dates**, for example when funds are needed or planned fundraising activities. These should also appear in your schedule (section 1).
- c) **Managing the budget**: how often you will review your budget and how you will respond to shortfalls or savings.

3. Project Technical Objectives

List and discuss the technical goals your team will aim for in the design and operation of your rover in order to win the competition. Typically, these are measures you want to minimise, maximise or keep within a limit. For example, you might choose to minimise mass, maximise autonomy, minimise bandwidth, or some combination of these or other goals. Your objectives give the whole team a shared technical direction.

4. Initial Technical Performance Measures (Technical Budgets)

A Technical Performance Measure (TPM) is any measurable technical quantity that is important to your design, difficult to achieve or risky to your project's success. TPMs often come from your Technical Objectives. Examples include total mass, communication bandwidth, energy use or speed.

Provide a table of TPMs including at least the items below. Discuss these only as needed.

- a) **Your chosen Technical Performance Measures.**
- b) **An initial target value** for each TPM to be achieved by the competition.
- c) **Allocation**: how each target value is divided across the elements of your early System Hierarchy (the parts should add up to the total).

- d) **Critical technical milestones and dates:** decision points in the design process where, if a target is not being met, you would change the design. These should also appear in your schedule (section 1).
- e) **Managing TPMs:** how often you will review your TPM values and how you will respond to shortfalls or better-than-expected performance.

Project Management Plan Guidance

Format

- Submit your PMP as a single PDF named PMP-Institution Name, with your institution's name on the plan.
- Maximum length is 5 pages. A cover page (signed by your Academic Supervisor) does not count towards the limit. Any content beyond 5 pages will not be assessed.
- Present it professionally and organise it so each required element is easy to find. Use correct spelling and grammar and sensible margins.
- Main text must be at least 12 point. Text in figures and tables must be at least 9 point.

Helpful resources

1. A NASA video series introducing the key project management and systems engineering tools and how to apply them: [Systems Engineering for University-level Engineering Projects and Competitions](#).
2. Seminar from the 2024 NASA Lunabotics competition: ["How to Develop the Perfect Lunabotics SE Paper"](#).
3. Seminar from the 2024 NASA Lunabotics competition: ["Systems 'd'esign"](#).

Tips for a strong PMP

The four sections in the Project Management Plan section are the criteria your PMP is assessed against. These tips come from previous Lunabotics competitions.

Tip 1: make it readable. If you use a Gantt chart or tables to cover an assessment criterion, make sure they can be read. If we can't read it, we can't give you credit. If a graphic contains text smaller than 9 point in the PDF, also cover the key information in the main text (at least 12 point).

Tip 2: show your major reviews. Make sure at least three reviews appear on your schedule: SRR, PDR and CDR. If you use different reviews, say which one replaces each.

Tip 3: a plan is decisions, written down. Developing any management plan comes down to making decisions and writing them down. Record those decisions at the start of the project and share them with your team. Update your plan as you learn more. The

assessment criteria cover an important subset of the decisions you will need to make, and ask you to tell us what you decided.

Tip 4: plan how your budget will evolve. As your design matures, you will learn how much the project will really cost. So that you don't run out of money, review your budget regularly. Decide at the start how you will handle a shortfall and what you will do with any surplus, and explain this in your PMP.

Tip 5: allocate your TPMs across the System Hierarchy. The System Hierarchy is the backbone of your project. At the start it may have only two levels, but that is enough to allocate time, money and TPM values (such as mass) to each part so the whole system can achieve the mission. As the design matures, these allocations guide you to deeper levels of the hierarchy, and you may need to revise them. There is an example of allocation in [SE Video 2: "The Central Elements of Project Management"](#). The video covers budget allocation, and the same approach works for mass and other TPMs.

Glossary

Term	Meaning
Academic Supervisor	The member of staff at your institution who supports your team and signs your Statement of Support.
Berm	A raised bank or wall of material. In this challenge, a pile of regolith simulant built in the target area.
CDR – Critical Design Review	A review held when the detailed design is complete, to confirm it is ready to build and test.
COTS	Commercial off-the-shelf: a standard product bought ready-made rather than designed by the team.
Gantt chart	A bar chart showing project tasks and milestones against time.
Plume ejecta	Dust and rock blasted outwards by a spacecraft's engines during landing and take-off.
PDR – Preliminary Design Review	A review of your chosen design concept, to confirm it can meet the requirements before detailed design begins.
PMP – Project Management Plan	Your plan for the project's schedule, budget and technical targets, submitted as your application.
Proof of Life	A short video showing your rover moving material and its emergency stop working, due 14 May 2027.
P&D – Presentation and Demonstration	A short presentation to the judges at the competition about your design and team.
Regolith / regolith simulant	Regolith is the loose soil and rock on the Moon's surface. A simulant is an Earth material used to imitate it.
Rulebook	The detailed rules for the arena, runs and scoring, published at uklunabotics.co.uk .
SRR – System Requirements Review	An early review to confirm that your team understands the mission and has a complete, agreed set of requirements for the system.
Systems Engineering Paper	An optional write-up of how your team applied systems engineering during the project.
System Hierarchy	A breakdown of your rover into subsystems and components (for example chassis, drive, excavation, power, control).

Term	Meaning
TPM – Technical Performance Measure	A measurable technical quantity you set a target for and track, such as mass or energy use.

Contact us

Email: uklunabotics@open.ac.uk Website: uklunabotics.co.uk

We look forward to receiving your application.